

Week 4 - In class Assignment - Gephi Exercise: Facebook Social Network Analysis

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Dataset: SNAP Facebook Social Circles Dataset
File Used: facebook_combined.txt.gz → converted to facebook_edges.csv

1. Introduction

This project analyzes the Facebook friendship network from the SNAP dataset. The dataset contains anonymized Facebook users and friendship connections between them. Each node represents a Facebook user, while each edge represents a friendship relation between two users.

2. Dataset Overview

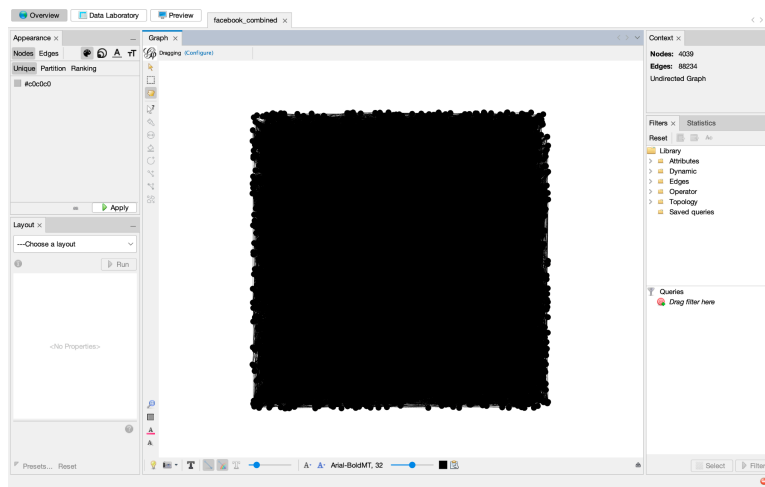
The dataset used was: facebook_combined.txt.gz. After importing the dataset into Gephi as an **Undirected Graph**, the following statistics were observed:

Metric	Value
Number of Nodes	4,039
Number of Edges	88,234
Graph Type	Undirected
Average Degree	43.691
Graph Density	0.011
Network Diameter	8
Average Path Length	3.693
Modularity Score	0.835

The network is highly connected but still divided into distinct communities.

3. Network Visualization

The network layout was generated using the **Yifan Hu Layout** in Gephi. This layout helped separate communities clearly and reduced node overlap. **Full Network Visualization:**



4. Degree Centrality Analysis

Degree centrality measures how many direct connections each node has.

Nodes with high degree centrality are considered highly influential or socially active users in the network. **Top 5 Nodes by Degree Centrality:**

Rank	Node ID	Degree
1	107	1045
2	1684	792
3	1912	755
4	3437	547
5	0	347

Interpretation:

- Node 107 is the most connected user in the network.
- These highly connected nodes likely represent central individuals with large friend circles.
- High-degree nodes significantly influence the connectivity of the network.

5. Betweenness Centrality Analysis

Betweenness centrality identifies nodes that act as bridges between communities.

These nodes are important because they connect otherwise separate groups. **Top 5 Nodes by Betweenness Centrality:**

Rank	Node ID	Betweenness Centrality
1	107	Highest
2	1684	Very High
3	1912	High
4	3437	High
5	0	Moderate

Interpretation:

- Node 107 appears to be both highly connected and strategically positioned.
- These nodes help information flow between different clusters.
- Removing bridge nodes could fragment the network into disconnected communities.

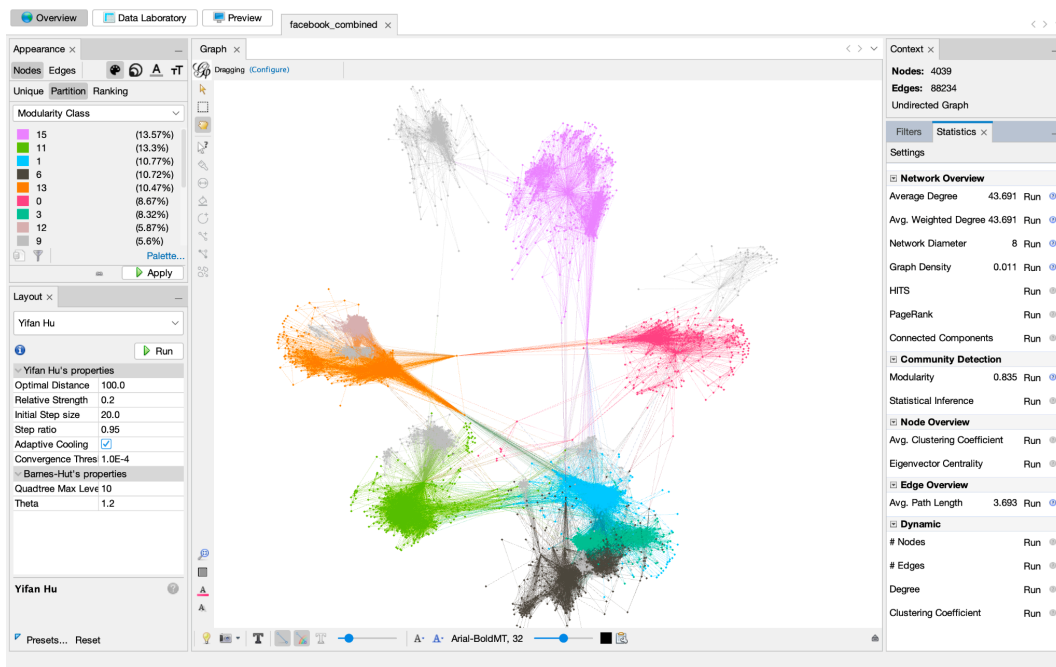
6. Community Detection (Modularity)

The **Modularity Algorithm** was used to identify communities inside the Facebook network.

The modularity score obtained was: 0.835

A high modularity score indicates strong community structure.

The network was colored based on **Modularity Class**, which revealed several distinct social groups.
Community Visualization:



Observation:

- Multiple communities are visible with clear separation.
- Each color represents a different friend group or social circle.
- Some communities are densely connected internally but weakly connected externally.

7. Degree Filter Analysis

A degree filter was applied: $\text{Degree} \geq 5$

This removed nodes with very few connections. **Comparison Between Original and Filtered Network:**

Original Network	Filtered Network
Contains all nodes	Removes weakly connected nodes
More cluttered visualization	Cleaner and easier to analyze
Includes peripheral users	Highlights core social structure
Dense overall appearance	Major communities become clearer

Interpretation:

- Filtering improved readability.
- Important communities and influential nodes became easier to identify.

- Peripheral nodes had limited impact on the core structure.

8. Conclusion

The SNAP Facebook dataset provides a strong example of how complex social interactions can be analyzed using network science.

The analysis revealed:

- influential users,
- bridge nodes,
- strong community structures,
- and highly interconnected social groups.

Using Gephi made it possible to visually and statistically understand the behavior of a large-scale social network.